

6. Vertebrate Neural Architecture

[FRAMING SENTENCE AND GUIDING QUESTIONS — authored last, after the Close, per the style guide's *Chapter opening*.]

A sea lamprey looks at first like an eel. It is long and slender, with no paired fins, and it swims by sending a wave down its whole body. The difference is at the front. A lamprey has no jaws. In place of a jaw, it has a round sucking disc lined with small, hard teeth. An adult sea lamprey feeds by fastening that disc to the side of a larger fish and drawing off its blood.

Lampreys and hagfishes are the only jawless fishes still alive. Their branch was the first to separate from the rest of the living vertebrates, more than 500 million years ago. That does not make the lamprey an ancestor of anything. It is a modern animal with a long history of its own. Even so, most of the parts that a human brain is organized around are already present in a lamprey's brain. One part is conspicuously missing.

Why those parts matter goes back to a problem we left open at the end of Module 5. We had described four kinds of learning. All four run on one principle, in the same kind of tissue, wherever there are synapses. A principle that general can be reused anywhere, and for that very reason it is specialized for nothing. A nervous system that does everything in one place cannot be very good at any one thing. And it cannot do two incompatible things at once.

Dividing the work requires something to divide. The nematode of Modules 4 and 5 has 302 neurons in its whole body, and its nervous system offers little to divide. An animal with vastly more neurons can set some of them aside for one job and others for another. Whether that is ever the better arrangement is not obvious. Separate parts need wiring between them. They also need some way of settling what happens when they disagree. So the first question is when separate structures are favored at all, and why (Table 6.1).

Table 6.1 — *Marr's three levels, the question each asks, the discipline that asks it, and this module's example.*

Level	The question it asks	Discipline	This module's example: a brain divided into parts
Computational	What problem is being solved, and why does solving it matter?	Evolution and ecology	Why an animal would build separate structures, rather than run one general mechanism everywhere
Algorithmic	What is represented, and by what procedure?	Cognitive psychology and AI	How the labor is divided: what each structure does differently, and how one action is chosen when several are proposed
Implementational	By what physical means, and at what cost?	Neuroscience	The parts of the vertebrate brain, and how an embryo builds them